

MATH 180 Strategies of Problem Solving

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September 10, 2025

Outline

- 1 SoPS Lecture 5
 - Battleship
 - Probability and Expectation Values
 - Wrapping up

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Chomp Review

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- we explored 3×3 Chomp

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- we learned **proof by cases**

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- we learned **proof by cases**
- we learned the **Strong Law of Small Numbers** and saw examples

Battleship Introduction



Activity (30 minutes)

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- review the rules of Battleship

Battleship Introduction



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- split into pairs and play some games!

Battleship Introduction



Activity (30 minutes)

- review the rules of Battleship
- split into pairs and play some games!
- can you observe any useful strategies?

Battleship Discussion

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Question 1

How does the game Battleship differ from other games we have played in this class. How is it similar?

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How does the game Battleship differ from other games we have played in this class. How is it similar?

Question 2

What mathematical tools could be useful in exploring the game Battleship?

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Probability Intro

Vocabulary

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- the set Ω of all possible outcomes is the **sample space**
- subsets $E \subseteq \Omega$ are called **events**
- the **probability** of an event E is

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|}$$

Probability Example

Rolling a Die

Experiment: roll a six-sided die, the sample space is:

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

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- the event of “getting a 3” is $\{3\}$

$$\mathbb{P}(\{3\}) = \frac{|\{3\}|}{|\Omega|} = \frac{1}{6}.$$

- the event of “getting an even number” is $\{2, 4, 6\}$

$$\mathbb{P}(\{2, 4, 6\}) = \frac{|\{2, 4, 6\}|}{|\Omega|} = \frac{3}{6} = \frac{1}{2}.$$

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Experiment: draw a card from a standard deck

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$$E = \{(J, \heartsuit), (Q, \heartsuit), (K, \heartsuit)\}$$

- what is the probability of this event?

$$P(E) = \frac{|E|}{|\Omega|} = \frac{3}{52}$$

Probability Meaning

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- record how many times the outcome is in E
- the fraction of times the outcome is in E should be close to $\mathbb{P}(E)$
- more experiments $\rightarrow$ closer to $\mathbb{P}(E)$

Probability Experiment

Activity (5 minutes)

Roll a die and keep track of

- how many rolls you do
- how many times the outcome is even

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Gather the data

Add up the results in your group and be prepared to share!

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Examples

- how much money we win/lose in a bet
- the position of a random walker

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Start with a sample space

$$\Omega = \{a_1, a_2, a_3, \dots, a_n\}.$$

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A **expectation value** of a **value function** $f(x)$ is

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Meaning

Represents the average value we obtain if we do thousands of experiments

Expectation Value Example

Gambling with dice

Suppose we play a game where we roll a die. If the outcome is a 6, then we win 4 dollars. Otherwise, we lose a dollar.

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- if we play a thousand games, we'll lose an average of 17

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Question 2] What is the probability of a ship being in each spot?

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Question 3] How many things are in the sample space?

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Question 4] What is the probability of a ship being in each spot?

Expectations in Battleship

Suppose we take our first shot at the center and we miss.

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Question 5] How do the probabilities change?

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Question 1

What does probability mean in your own words?

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Question 1

What does probability mean in your own words?

Question 2

How can we use probability to play Battleship better?

Question of the Day

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Puzzle

Suppose we play mini-Battleship on a 5×5 board. How many ways are there to place a 2×1 Destroyer and a 3×1 submarine on the board **simultaneously**.

Final thoughts

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- Can you think of any other games where probability may be an important tool? Describe one.
- What strategies would you suggest for placing the ships in Battleship? What about firing the shots?

Submit your responses by **Tonight!**